

A Multi-Factor Intelligent Major Recommendation System

Decision Support Using Logical, Emotional, and Psychological Analysis

Course: Artificial Intelligence (CS3081)

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Abstract

Selecting an academic major is a complex decision influenced by academic performance, interests, personality, and job market trends. This project presents an intelligent recommendation system that integrates logical, emotional, and psychological factors using a weighted model. The system generates ranked recommendations, provides explanations, and stores feedback, improving transparency and decision-making. In addition, the system incorporates input validation mechanisms to ensure data consistency and reliability. By combining multiple dimensions with explainable outputs, the proposed solution enhances user trust and supports more informed academic decision-making.

1 Introduction

Choosing a university major is a critical step that shapes a student's academic path and career opportunities. Many students lack guidance and rely on incomplete information. This project introduces an intelligent system that analyzes multiple factors to provide personalized recommendations. Traditional advising methods often overlook the complexity of student preferences and abilities. Therefore, there is a growing need for intelligent systems that can analyze multiple dimensions simultaneously. This project addresses this need by proposing a multi-factor recommendation system that combines both quantitative and qualitative aspects of decision-making.

2 Problem Statement

Students often choose majors based on limited or biased information, leading to dissatisfaction and poor performance. This project aims to build a system that evaluates multiple aspects of a student's profile and provides clear recommendations. Moreover, the lack of personalized guidance can lead to mismatches between students and their chosen fields, resulting in decreased motivation and performance. This highlights the importance of developing systems that can adapt to individual differences and provide tailored recommendations.

3 Objectives

- Develop an intelligent recommendation system
- Integrate logical, emotional, and psychological factors

- Generate top three ranked recommendations
 - Provide explainable outputs
 - Implement feedback mechanism
 - Ensure robustness against inconsistent inputs
- These objectives collectively aim to build a system that is not only accurate, but also transparent, user-friendly, and robust against invalid inputs.

4 Methodology

The system uses a weighted scoring model:

$$Score = 0.40(Logical) + 0.30(Emotional) + 0.20(Psychological) + 0.10(Market)$$

Each major is evaluated based on the student profile, and the highest scores determine recommendations. Each component in the model represents a different aspect of the student's profile. The logical component evaluates academic abilities, the emotional component reflects personal interests, and the psychological component captures personality traits. The market factor ensures that recommendations align with real-world demand.

This multi-factor approach ensures a balanced evaluation, avoiding bias toward a single dimension and producing more reliable recommendations.

5 Implementation

The system was implemented using Python and Streamlit. The system architecture consists of several interconnected modules, including data input, validation, scoring, recommendation generation, and feedback stor-

age. Streamlit was used to provide an interactive user interface, allowing real-time interaction with the system.

5.1 User Interface

A Multi-Factor Intelligent Major Recommendation System
Personalized major recommendations using logical, emotional, psychological, and job market factors.

Student Profile

Student	GPA Level	Math	Top Interest
Noora	High	Medium	Technology
GPA	MBTI	Programming	Engagement
4.5	INTJ	Medium	Medium

▼ View full student profile

	Student_ID	GPA	GPA_Level	Math_Skill	Programming_Skill	ProblemSolving	AnalyticalThinking	Engagement	Interest_Tech	Interest_Business	Interest_Design	Interest_Con
0	Noora	4.5	High	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium

Figure 1: User interface displaying student input and profile

6 Results

The system generates three ranked recommendations. The results demonstrate the system's ability to generate consistent and meaningful recommendations based on different student profiles. The ranking mechanism ensures that the most suitable majors are presented clearly, along with their corresponding scores.

6.1 Top Recommendations

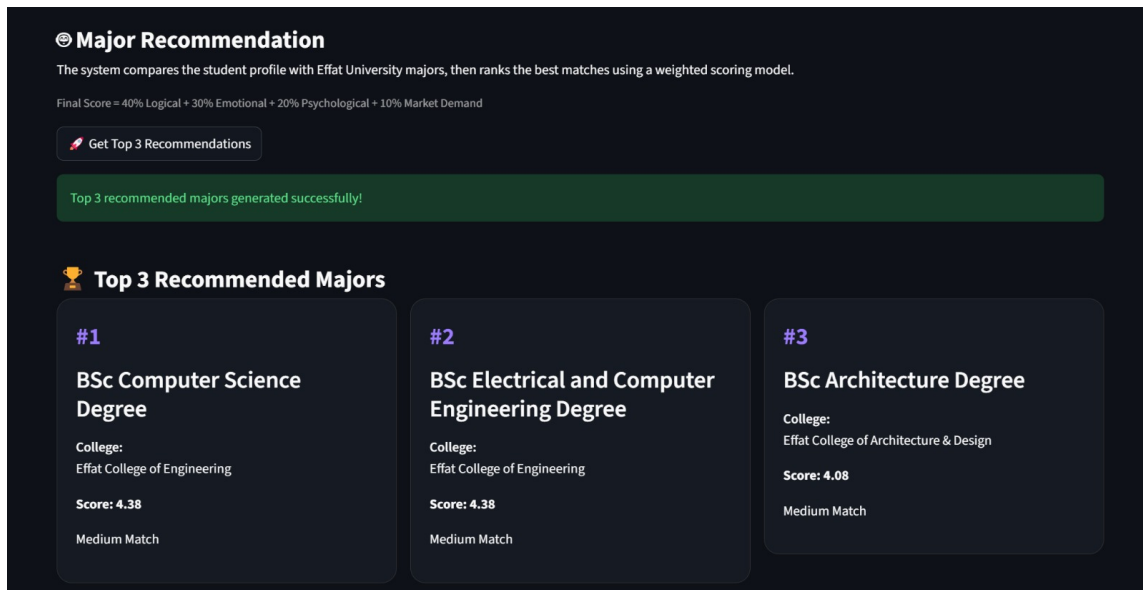


Figure 2: Top three recommended majors with scores

6.2 Explanation

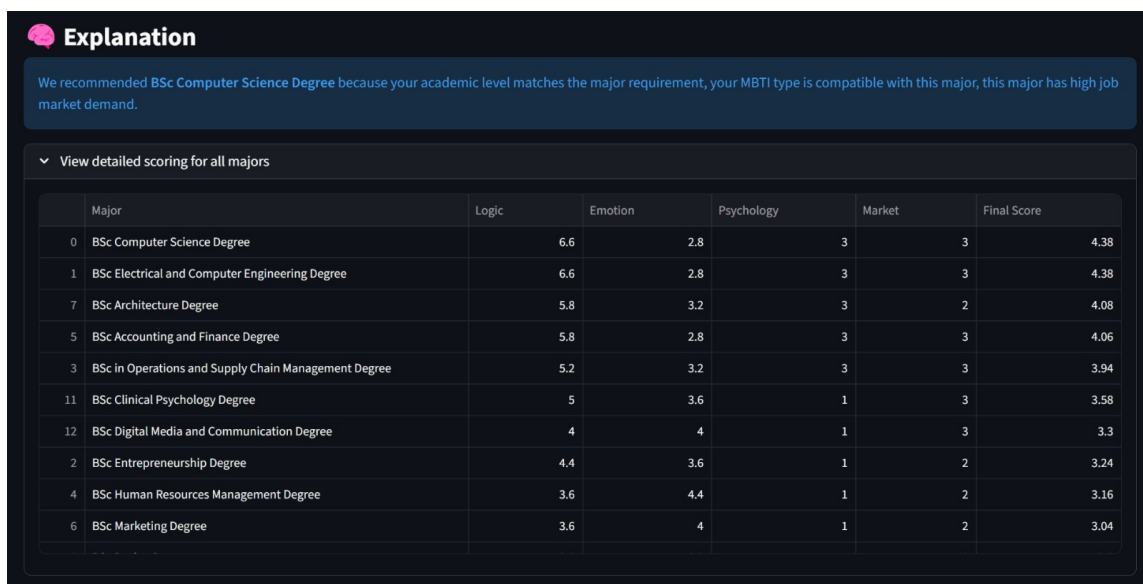


Figure 3: Explanation of the top recommendation

7 Robustness and Validation

The system includes validation checks to detect inconsistent inputs. This module plays a critical role in maintaining the integrity of the system by preventing invalid or contradictory inputs from affecting the results. By enforcing validation rules, the system ensures that only reliable data is used during the recommendation process.

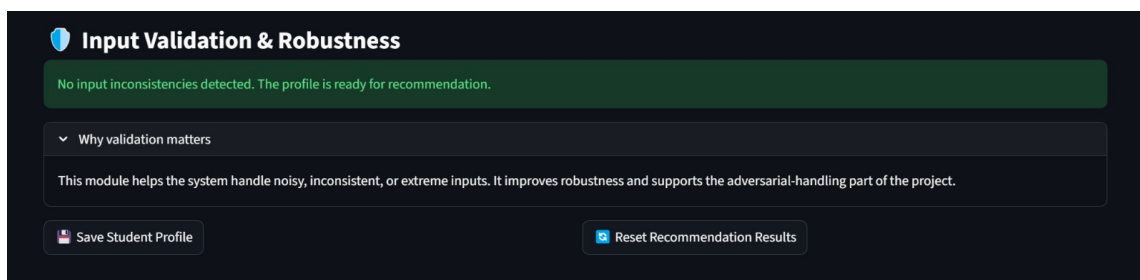


Figure 4: Validation module ensuring input consistency

8 User Feedback

The system allows users to select a final major. This feedback mechanism allows the system to capture user preferences, which can be utilized in future enhancements such as adaptive learning or personalized model tuning.



Figure 5: User feedback interface storing final decision

9 Discussion

The developed system demonstrates the effectiveness of integrating multiple dimensions in decision-making, including academic performance, per-

sonal interests, and personality traits. Unlike traditional approaches that rely on a single factor, this system combines logical, emotional, and psychological components to provide more balanced and meaningful recommendations.

Additionally, the explanation module enhances transparency by clearly justifying the recommended major, allowing users to understand the reasoning behind each decision. This contributes to building user trust and confidence in the system.

The validation mechanism further strengthens the system by detecting inconsistent or unrealistic inputs, ensuring that recommendations are generated only from reliable data. Overall, the system highlights the importance of combining intelligent analysis with user-centered design in AI applications.

10 Conclusion

This project presents a multi-factor intelligent major recommendation system that supports students in making informed academic decisions. By integrating logical, emotional, psychological, and market-based factors, the system provides personalized and well-structured recommendations.

Furthermore, the inclusion of explainability and input validation enhances both the reliability and usability of the system. The results demonstrate that combining multiple perspectives leads to more accurate and meaningful recommendations compared to traditional methods.

Overall, this system not only assists in selecting a suitable major but also promotes a deeper understanding of the factors influencing the decision, making it a valuable tool for academic planning.

11 Future Work

Future improvements to this system may include the integration of real-world datasets to enhance accuracy and reliability. In addition, incorporating machine learning techniques could allow the system to learn from user feedback and continuously improve its recommendations.

Expanding the majors database to include a wider range of academic programs and career paths would further increase the system's applicability. Finally, deploying the system as a scalable web application could enable broader access for students and educational institutions.

At its core, the system is not just about selecting a major, but about understanding the student behind the decision.

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